

ZEHAO SHEN

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■ EDUCATION

Zhejiang University

Sept. 2023 – Jun. 2027 (Expected)

B.E. in Information Engineering, College of Information Science and Electronic Engineering

Overall GPA: 4.09/4.3

■ RESEARCH INTERESTS

My research interests center on **LLM agents and reasoning for scientific and engineering problems**, particularly their applications to **wireless communications**. I am also interested in how such intelligent systems can be efficiently deployed through **embedded and edge AI**, with a broader focus on **hardware–software co-design** for practical AI systems.

■ PUBLICATIONS

Radio-Scene-Native Digital Twin Datasets for Wireless Communication: Generation, Simulation, and Validation

Co-author · Manuscript in progress

■ RESEARCH & INNOVATION EXPERIENCE

Taxonomy of LLM Reasoning Architectures

Jul. 2026 – Present

Nanyang Technological University, Singapore

Visiting Researcher | Advisor: Prof. Chau Yuen

- Developed a systematic **taxonomy of LLM reasoning architectures** through open coding of 18 representative systems, characterizing how reasoning mechanisms structure computation, incorporate feedback, and maintain state; validated coding stability across independent annotations with Fleiss' κ of **0.87–0.97**.
- Designed and executed controlled comparisons of Direct prompting, PAL, Zero-shot CoT, and Self-Refine across GSM1K, HARP-small, MATH-Perturb, and U-MATH, covering **3,608 unique problems** and over **28K evaluation records** under standardized model, sampling, and repeat settings.
- Identified substantial task-dependent variation in architectural effectiveness: in completed controlled experiments, Direct prompting exceeded PAL by **7.58 pp** overall, while the performance gap varied from **1.61 pp** to **18.80 pp** across datasets, motivating broader validation of the taxonomy across reasoning mechanisms and tasks.

WirelessAgent++ Reproduction & Workflow Evaluation

Jul. 2026

Nanyang Technological University, Singapore

Visiting Researcher | Advisor: Prof. Chau Yuen

- Reproduced and extended the experimental pipeline of **WirelessAgent++**, evaluating modular agent operators, feedback mechanisms, and MCTS-based workflow search across **7 optimizer–executor configurations**, **1,412 unique problems**, and over **17K valid model evaluations**.
- Built a reproducible and auditable evaluation workflow with resumable execution, experiment manifests, API/token telemetry, and partial-credit scoring.
- Audited and improved the **WCHW scoring methodology**, identifying systematic issues that materially affected workflow rankings; validated the revised scorer on **5,728 previously generated outputs**.

Telecom AI Dataset Construction

Zhejiang University, Hangzhou, China

Mar. 2026 – Jun. 2026

Undergraduate Researcher | Advisor: Prof. Chongwen Huang

- Contributed to reasoning-data construction for **TelecomGPT-R1**, a telecom reasoning model trained on a corpus of over **158K examples**, with a focus on mathematical derivation and numerical computation problems.
- Worked on data-processing pipelines that convert textbooks, lecture notes, papers, and open problem sets into self-contained **problem–reasoning–answer** samples through problem filtering, decomposition, reasoning-chain generation, and sample augmentation.
- Participated in **multi-stage verification** using numerical consistency checks, unit-aware comparison, symbolic equivalence, Python recomputation, and independent re-solving to improve sample correctness and reproducibility.

EDU-Mate: Multimodal Intelligent Learning Assistant

Zhejiang University, Hangzhou, China

Mar. 2026 – Jul. 2026

Agent & Software Lead | 2026 Intel Cup

- Developed an end-to-end multimodal classroom assistant integrating real-time speech transcription, visual understanding, knowledge graphs, RAG, and source-grounded LLM Agent workflows for in-class assistance and post-class review.
- Designed the core Agent orchestration and system integration across a FastAPI backend, React/TypeScript frontend, WebSocket event pipeline, and hybrid edge–cloud inference; deployed **WhisperLive/OpenVINO** on iGPU/GPU and **Qwen2.5-3B-INT4** via OpenVINO GenAI on CPU.
- Validated the system in an end-to-end classroom setting, continuously processing **171 transcript segments**, persisting stable transcripts within **5 s**, and generating post-class summaries and to-do items within **5 min**.

5-Stage Pipelined RISC-V CPU

Zhejiang University

Oct. 2025 – Dec. 2025

Sole Designer & Developer | Independent Project

- Designed and implemented a **32-bit, 5-stage pipelined RISC-V CPU** in Verilog, supporting 32 required RV32I instructions and extending the memory subsystem with 6 byte/half-word load-store instructions.
- Implemented **MEM/WB-to-EX forwarding**, register-file bypassing, and a **Hazard Detection Unit** with one-cycle Load-Use stalls; added early branch resolution and pipeline flushing to handle control hazards.
- Verified ALU, decode, fetch, and full-pipeline behavior with module- and system-level testbenches, then synthesized and deployed the processor on a **Nexys Video FPGA**.

Closed-Loop Temperature Control System

Zhejiang University

Jun. 2025 – Jul. 2025

System Designer | Schematic, Control Algorithm & Hardware Integration

- Designed the schematic and hardware around an **STM32**, integrating NTC/ADC and I2C GY906 sensing, a MOSFET-PWM heater stage, and a custom PCB; measured up to 13 W heating power with a stable 5.04 V regulated rail.
- Implemented and tuned a **gain-scheduled PID controller** with sliding-window filtering and PWM actuation, meeting all control targets: $\pm 2^{\circ}\text{C}$ steady-state regulation, $< 2^{\circ}\text{C}$ overshoot, and $< 10\text{ s}$ settling time.
- Completed hardware-software integration through oscilloscope-based PWM verification and NTC curve calibration, and implemented OLED interaction, over-temperature alarms, and **UART/Bluetooth** remote control.

■ TECHNICAL STRENGTHS

Programming Language	C/C++, Python, MATLAB, Verilog TOEFL after January 21, 2026: 5.0 (R:6, L:6, W:4, S:4.5), Mandarin (Native)
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■ HONORS & AWARDS

Third Prize, Intel Cup Embedded System Contest, National Undergraduate Electronic Contest	Jul. 2026
Zhejiang Provincial Government Scholarship, Zhejiang Provincial Government	Nov. 2025
Outstanding Student for Academic Excellence, Zhejiang University (Awarded twice)	Oct. 2024 & 2025